```
Last login: Thu Sep 17 17:12:08 on ttys008
(base) zainabsiddiqui@Zainabs-MacBook-Pro synteny % conda create -n prdm9_synten
y -c conda-forge -c bioconda ncbi-datasets-cli blast biopython
Channels:
 - conda-forge
 - bioconda
 - defaults
Platform: osx-arm64
Collecting package metadata (repodata.json): -
done
Solving environment: done

## Package Plan ##

  environment location: /opt/anaconda3/envs/prdm9_synteny

  added / updated specs:
    - biopython
    - blast
    - ncbi-datasets-cli


The following packages will be downloaded:

    package                    |            build
    ---------------------------|-----------------
    _openmp_mutex-4.5          |       8_kmp_llvm           8 KB  conda-forge
    biopython-1.88             |  py314h6dfba8d_1         3.2 MB  conda-forge
    blast-2.17.0               |       h81fc81c_1       135.7 MB  bioconda
    bzip2-1.0.8                |      h4e30115_10         122 KB  conda-forge
    c-ares-1.34.8              |       h74c22ad_2         193 KB  conda-forge
    ca-certificates-2026.7.22  |       hbd8a1cb_0         129 KB  conda-forge
    curl-8.22.0                |       h6651222_0         178 KB  conda-forge
    entrez-direct-26.0         |       h2869703_0        14.4 MB  bioconda
    icu-78.3                   |  py310h579977c_2        13.4 MB  conda-forge
    krb5-1.22.2                |       h34f8a20_2         1.1 MB  conda-forge
    libblas-3.11.0             |11_h51639a9_openblas          18 KB  conda-forge
    libcblas-3.11.0            |11_hb0561ab_openblas          18 KB  conda-forge
    libcurl-8.22.0             |       h6651222_0         410 KB  conda-forge
    libcxx-23.1.2              |       h55c6f16_0         566 KB  conda-forge
    libedit-3.1.20250104       | pl5321h26f1114_1         105 KB  conda-forge
    libev-4.33                 |       h1a92334_3          39 KB  conda-forge
    libexpat-2.8.5             |       h6111c0a_0          68 KB  conda-forge
    libffi-3.7.0               |       h47dc5ef_1          43 KB  conda-forge
    libgcc-16.2.0              |       h3cf6597_7         357 KB  conda-forge
    libgfortran-16.2.0         |       h07b0088_7          97 KB  conda-forge
    libgfortran5-16.2.0        |       hdb7a957_7         542 KB  conda-forge
    libiconv-1.18              |       he4c29f2_3         733 KB  conda-forge
    libidn2-2.3.8              |       ha90df94_1         142 KB  conda-forge
    libintl-0.25.1             |       hae62183_2          88 KB  conda-forge
    liblapack-3.11.0           |11_hd9741b5_openblas          18 KB  conda-forge
    liblzma-5.8.3              |       h8088a28_1          90 KB  conda-forge
    libmpdec-4.0.0             |       h84a0fba_2          72 KB  conda-forge
    libnghttp2-1.68.1          |       h435687b_1         554 KB  conda-forge
    libopenblas-0.3.34         |openmp_h5e6e99c_1         4.1 MB  conda-forge
    libpsl-0.23.1              |       hdb0c161_1          71 KB  conda-forge
```

```
    libpython-3.14.7           |he655c36_6_cp314t         1.9 MB  conda-forge
    libsqlite-3.53.4           |       hca69786_1         921 KB  conda-forge
    libssh2-1.11.1             |       hbf2880b_1         274 KB  conda-forge
    libunistring-0.9.10        |       h3422bc3_0         1.5 MB  conda-forge
    libzlib-1.3.2              |       h8088a28_3          47 KB  conda-forge
    llvm-openmp-23.1.2         |       hdb3d66b_0         281 KB  conda-forge
    ncbi-datasets-cli-18.37.0  |       hce30654_0        32.3 MB  conda-forge
    ncbi-vdb-3.4.1             |       hfc726f9_0         4.5 MB  bioconda
    ncurses-6.6                |       he64c551_1         785 KB  conda-forge
    numpy-2.5.3                |  py314hceca211_0         7.0 MB  conda-forge
    openssl-3.6.4              |       h55eecbc_0         3.0 MB  conda-forge
    pcre2-10.47                |       he63d830_1         832 KB  conda-forge
    perl-5.32.1                |10_hb001647_perl5        11.7 MB  conda-forge
    perl-archive-tar-3.12      | pl5321hdfd78af_0          38 KB  bioconda
    perl-carp-1.50             | pl5321hd8ed1ab_0          22 KB  conda-forge
    perl-common-sense-3.75     | pl5321hd8ed1ab_0          20 KB  conda-forge
    perl-compress-raw-bzip2-2.214| pl5321h2ddc596_0          53 KB  conda-forge
    perl-compress-raw-zlib-2.214| pl5321ha1c2b25_0          73 KB  conda-forge
    perl-encode-3.24           | pl5321h84a0fba_0         1.0 MB  conda-forge
    perl-exporter-5.74         | pl5321hd8ed1ab_0          19 KB  conda-forge
    perl-exporter-tiny-1.002002| pl5321hd8ed1ab_0          28 KB  conda-forge
    perl-extutils-makemaker-7.70| pl5321hd8ed1ab_0         154 KB  conda-forge
    perl-io-compress-2.216     | pl5321haef7865_0          86 KB  bioconda
    perl-io-zlib-1.15          | pl5321hdfd78af_1          12 KB  bioconda
    perl-json-4.11             | pl5321hdfd78af_0          52 KB  bioconda
    perl-json-xs-4.04          | pl5321h4675bf2_0          67 KB  bioconda
    perl-list-moreutils-0.430  | pl5321hdfd78af_0          32 KB  bioconda
    perl-list-moreutils-xs-0.430| pl5321hbdacb55_5          42 KB  bioconda
    perl-parent-0.243          | pl5321hd8ed1ab_0          14 KB  conda-forge
    perl-pathtools-3.75        | pl5321hc71e825_2          48 KB  conda-forge
    perl-scalar-list-utils-1.70| pl5321h44e845a_0          46 KB  conda-forge
    perl-storable-3.15         | pl5321hc71e825_2          62 KB  conda-forge
    perl-types-serialiser-1.01 | pl5321hdfd78af_0          13 KB  bioconda
    pip-26.2.1                 |     pyh145f28c_0         1.1 MB  conda-forge
    python-3.14.7              |h6f66cd8_6_cp314t        13.8 MB  conda-forge
    python_abi-3.14            |         9_cp314t           7 KB  conda-forge
    readline-8.3               |       h8b90a29_1         307 KB  conda-forge
    tk-8.6.13                  |       hbeba79b_4         3.2 MB  conda-forge
    tzdata-2026c               |       h151e31d_0         116 KB  conda-forge
    wget-1.25.0                |       h7ba78b2_1         318 KB  conda-forge
    zlib-1.3.2                 |       h8088a28_3          80 KB  conda-forge
    zstd-1.5.7                 |       hf451053_7         424 KB  conda-forge
    ------------------------------------------------------------
                                           Total:       262.6 MB

The following NEW packages will be INSTALLED:

  _openmp_mutex      conda-forge/osx-arm64::_openmp_mutex-4.5-8_kmp_llvm
  biopython          conda-forge/osx-arm64::biopython-1.88-py314h6dfba8d_1
  blast              bioconda/osx-arm64::blast-2.17.0-h81fc81c_1
  bzip2              conda-forge/osx-arm64::bzip2-1.0.8-h4e30115_10
  c-ares             conda-forge/osx-arm64::c-ares-1.34.8-h74c22ad_2
  ca-certificates    conda-forge/noarch::ca-certificates-2026.7.22-hbd8a1cb_0
  curl               conda-forge/osx-arm64::curl-8.22.0-h6651222_0
  entrez-direct      bioconda/osx-arm64::entrez-direct-26.0-h2869703_0
  icu                conda-forge/osx-arm64::icu-78.3-py310h579977c_2
```

```
  krb5               conda-forge/osx-arm64::krb5-1.22.2-h34f8a20_2
  libblas            conda-forge/osx-arm64::libblas-3.11.0-11_h51639a9_openblas
  libcblas           conda-forge/osx-arm64::libcblas-3.11.0-11_hb0561ab_openblas

  libcurl            conda-forge/osx-arm64::libcurl-8.22.0-h6651222_0
  libcxx             conda-forge/osx-arm64::libcxx-23.1.2-h55c6f16_0
  libedit            conda-forge/osx-arm64::libedit-3.1.20250104-pl5321h26f1114_
1
  libev              conda-forge/osx-arm64::libev-4.33-h1a92334_3
  libexpat           conda-forge/osx-arm64::libexpat-2.8.5-h6111c0a_0
  libffi             conda-forge/osx-arm64::libffi-3.7.0-h47dc5ef_1
  libgcc             conda-forge/osx-arm64::libgcc-16.2.0-h3cf6597_7
  libgfortran        conda-forge/osx-arm64::libgfortran-16.2.0-h07b0088_7
  libgfortran5       conda-forge/osx-arm64::libgfortran5-16.2.0-hdb7a957_7
  libiconv           conda-forge/osx-arm64::libiconv-1.18-he4c29f2_3
  libidn2            conda-forge/osx-arm64::libidn2-2.3.8-ha90df94_1
  libintl            conda-forge/osx-arm64::libintl-0.25.1-hae62183_2
  liblapack          conda-forge/osx-arm64::liblapack-3.11.0-11_hd9741b5_openbla
s
  liblzma            conda-forge/osx-arm64::liblzma-5.8.3-h8088a28_1
  libmpdec           conda-forge/osx-arm64::libmpdec-4.0.0-h84a0fba_2
  libnghttp2         conda-forge/osx-arm64::libnghttp2-1.68.1-h435687b_1
  libopenblas        conda-forge/osx-arm64::libopenblas-0.3.34-openmp_h5e6e99c_1

  libpsl             conda-forge/osx-arm64::libpsl-0.23.1-hdb0c161_1
  libpython          conda-forge/osx-arm64::libpython-3.14.7-he655c36_6_cp314t
  libsqlite          conda-forge/osx-arm64::libsqlite-3.53.4-hca69786_1
  libssh2            conda-forge/osx-arm64::libssh2-1.11.1-hbf2880b_1
  libunistring       conda-forge/osx-arm64::libunistring-0.9.10-h3422bc3_0
  libzlib            conda-forge/osx-arm64::libzlib-1.3.2-h8088a28_3
  llvm-openmp        conda-forge/osx-arm64::llvm-openmp-23.1.2-hdb3d66b_0
  ncbi-datasets-cli  conda-forge/osx-arm64::ncbi-datasets-cli-18.37.0-hce30654_0

  ncbi-vdb           bioconda/osx-arm64::ncbi-vdb-3.4.1-hfc726f9_0
  ncurses            conda-forge/osx-arm64::ncurses-6.6-he64c551_1
  numpy              conda-forge/osx-arm64::numpy-2.5.3-py314hceca211_0
  openssl            conda-forge/osx-arm64::openssl-3.6.4-h55eecbc_0
  pcre2              conda-forge/osx-arm64::pcre2-10.47-he63d830_1
  perl               conda-forge/osx-arm64::perl-5.32.1-10_hb001647_perl5
  perl-archive-tar   bioconda/noarch::perl-archive-tar-3.12-pl5321hdfd78af_0
  perl-carp          conda-forge/noarch::perl-carp-1.50-pl5321hd8ed1ab_0
  perl-common-sense  conda-forge/noarch::perl-common-sense-3.75-pl5321hd8ed1ab_0

  perl-compress-raw~ conda-forge/osx-arm64::perl-compress-raw-bzip2-2.214-pl5321
h2ddc596_0
  perl-compress-raw~ conda-forge/osx-arm64::perl-compress-raw-zlib-2.214-pl5321h
a1c2b25_0
  perl-encode        conda-forge/osx-arm64::perl-encode-3.24-pl5321h84a0fba_0
  perl-exporter      conda-forge/noarch::perl-exporter-5.74-pl5321hd8ed1ab_0
  perl-exporter-tiny conda-forge/noarch::perl-exporter-tiny-1.002002-pl5321hd8ed
1ab_0
  perl-extutils-mak~ conda-forge/noarch::perl-extutils-makemaker-7.70-pl5321hd8e
d1ab_0
  perl-io-compress   bioconda/osx-arm64::perl-io-compress-2.216-pl5321haef7865_0

  perl-io-zlib       bioconda/noarch::perl-io-zlib-1.15-pl5321hdfd78af_1
```

```
  perl-json          bioconda/noarch::perl-json-4.11-pl5321hdfd78af_0
  perl-json-xs       bioconda/osx-arm64::perl-json-xs-4.04-pl5321h4675bf2_0
  perl-list-moreuti~ bioconda/noarch::perl-list-moreutils-0.430-pl5321hdfd78af_0

  perl-list-moreuti~ bioconda/osx-arm64::perl-list-moreutils-xs-0.430-pl5321hbda
cb55_5
  perl-parent        conda-forge/noarch::perl-parent-0.243-pl5321hd8ed1ab_0
  perl-pathtools     conda-forge/osx-arm64::perl-pathtools-3.75-pl5321hc71e825_2

  perl-scalar-list-~ conda-forge/osx-arm64::perl-scalar-list-utils-1.70-pl5321h4
4e845a_0
  perl-storable      conda-forge/osx-arm64::perl-storable-3.15-pl5321hc71e825_2
  perl-types-serial~ bioconda/noarch::perl-types-serialiser-1.01-pl5321hdfd78af_
0
  pip                conda-forge/noarch::pip-26.2.1-pyh145f28c_0
  python             conda-forge/osx-arm64::python-3.14.7-h6f66cd8_6_cp314t
  python_abi         conda-forge/noarch::python_abi-3.14-9_cp314t
  readline           conda-forge/osx-arm64::readline-8.3-h8b90a29_1
  tk                 conda-forge/osx-arm64::tk-8.6.13-hbeba79b_4
  tzdata             conda-forge/noarch::tzdata-2026c-h151e31d_0
  wget               conda-forge/osx-arm64::wget-1.25.0-h7ba78b2_1
  zlib               conda-forge/osx-arm64::zlib-1.3.2-h8088a28_3
  zstd               conda-forge/osx-arm64::zstd-1.5.7-hf451053_7


Proceed ([y]/n)?

Downloading and Extracting Packages:

Preparing transaction: done
Verifying transaction: done
Executing transaction: done
#
# To activate this environment, use
#
#     $ conda activate prdm9_synteny
#
# To deactivate an active environment, use
#
#     $ conda deactivate

(base) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(base) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(base) zainabsiddiqui@Zainabs-MacBook-Pro synteny % conda activate prdm9_synteny

(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % datasets --version
blastp -version
python --version
datasets version: 18.37.0
blastp: 2.17.0+
 Package: blast 2.17.0, build Aug 22 2026 11:50:41
Python 3.14.7
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
```

```
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % mkdir -p ./{turtle,
alligator,blast,regions,notes}
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % ls
alligator       blast           notes           regions         turtle
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % datasets summary ge
ne gene-id 101947848 --as-json-lines > turtle_prdm9.jsonl
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % cat turtle_prdm9.js
onl
{"alternate_names":["histone-lysine N-methyltransferase PRDM9"],"annotations":[{
"annotation_name":"GCF_011386835.1-RS_2024_05","annotation_release_date":"2024-0
5-29","assembly_accession":"GCF_011386835.1","assembly_name":"ASM1138683v2","gen
omic_locations":[{"genomic_accession_version":"NC_088810.1","genomic_range":{"be
gin":"992989","end":"1017260","orientation":"plus"},"sequence_name":"20"}]}],"ch
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omain 9","gene_groups":[{"id":"56979","method":"NCBI Ortholog"}],"gene_id":"1019
47848","gene_ontology":{"assigned_by":"RefSeq","biological_processes":[{"evidenc
e_code":"IEA","go_id":"GO:0006357","name":"regulation of transcription by RNA po
lymerase II","qualifier":"involved_in"}],"cellular_components":[{"evidence_code"
:"IEA","go_id":"GO:0005634","name":"nucleus","qualifier":"located_in"}],"molecul
ar_functions":[{"evidence_code":"IEA","go_id":"GO:0000981","name":"DNA-binding t
ranscription factor activity, RNA polymerase II-specific","qualifier":"enables"}
,{"evidence_code":"IEA","go_id":"GO:0000978","name":"RNA polymerase II cis-regul
atory region sequence-specific DNA binding","qualifier":"enables"},{"evidence_co
de":"IEA","go_id":"GO:0042054","name":"histone methyltransferase activity","qual
ifier":"enables"},{"evidence_code":"IEA","go_id":"GO:0005515","name":"protein bi
nding","qualifier":"enables"},{"evidence_code":"IEA","go_id":"GO:0008270","name"
:"zinc ion binding","qualifier":"enables"}]},"orientation":"plus","protein_count
":7,"symbol":"PRDM9","tax_id":"8478","taxname":"Chrysemys picta bellii","transcr
ipt_count":7,"transcript_type_counts":[{"count":7,"type":"PROTEIN_CODING_MODEL"}
],"type":"PROTEIN_CODING"}
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % datasets summary ge
ne gene-id 101947848 --as-json-lines | \
dataformat tsv gene \
--fields symbol,gene-id,tax-name,orientation,ref-standard-gene-range-accession,r
ef-standard-gene-range-range-start,ref-standard-gene-range-range-stop,transcript
-count,protein-count
Symbol  NCBI GeneID     Taxonomic Name  Orientation     Reference Standard Gene
Range Sequence Accession        Reference Standard Gene Range Start     Referenc
e Standard Gene Range Stop      Transcripts     Proteins
```

```
PRDM9   101947848       Chrysemys picta bellii  plus                            7
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % nano notes/turtle_g
ene_info.txt
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % nano notes/turtle_g
ene_info.txt
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % datasets download g
ene gene-id 101947848 \
    --include gene,rna,cds,protein,product-report \
    --filename turtle_prdm9.zip
/ Collecting gene records    1 record
Downloading: turtle_prdm9.zip    25.8kB valid data package
Validating package files [================================================] 100%
 9/9
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % unzip -o turtle_prd
m9.zip -d turtle/prdm9
Archive:  turtle_prdm9.zip
  inflating: turtle/prdm9/README.md
  inflating: turtle/prdm9/ncbi_dataset/data/gene.fna
  inflating: turtle/prdm9/ncbi_dataset/data/rna.fna
  inflating: turtle/prdm9/ncbi_dataset/data/cds.fna
  inflating: turtle/prdm9/ncbi_dataset/data/protein.faa
  inflating: turtle/prdm9/ncbi_dataset/data/data_report.jsonl
  inflating: turtle/prdm9/ncbi_dataset/data/product_report.jsonl
  inflating: turtle/prdm9/ncbi_dataset/data/dataset_catalog.json
  inflating: turtle/prdm9/md5sum.txt
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % find turtle/prdm9 -
type f
turtle/prdm9/ncbi_dataset/data/cds.fna
turtle/prdm9/ncbi_dataset/data/data_report.jsonl
turtle/prdm9/ncbi_dataset/data/rna.fna
turtle/prdm9/ncbi_dataset/data/gene.fna
turtle/prdm9/ncbi_dataset/data/protein.faa
turtle/prdm9/ncbi_dataset/data/product_report.jsonl
turtle/prdm9/ncbi_dataset/data/dataset_catalog.json
turtle/prdm9/README.md
```

```
turtle/prdm9/md5sum.txt
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % datasets summary ge
ne gene-id 101947848 \
    --ortholog "Alligator mississippiensis" \
    --as-json-lines
Warning: No gene orthologs found for the specified NCBI GeneID and taxon.
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % datasets download g
enome accession GCF_011386835.1 \
    --include gff3,protein \
    --filename turtle_annotation.zip
\ Collecting genome records    1 record
Downloading: turtle_annotation.zip    44.3MB valid data package
Validating package files [================================================] 100%
 6/6
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % datasets download g
enome accession GCA_030867095.1 \
    --include gff3,protein \
    --filename alligator_annotation.zip
| Collecting genome records    1 record
Downloading: alligator_annotation.zip    3.83kB valid data package
Validating package files [================================================] 100%
 4/4
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % unzip -o turtle_ann
otation.zip -d turtle/genome
unzip -o alligator_annotation.zip -d alligator/genome
Archive:  turtle_annotation.zip
  inflating: turtle/genome/README.md
  inflating: turtle/genome/ncbi_dataset/data/assembly_data_report.jsonl
  inflating: turtle/genome/ncbi_dataset/data/GCF_011386835.1/genomic.gff
  inflating: turtle/genome/ncbi_dataset/data/GCF_011386835.1/protein.faa
  inflating: turtle/genome/ncbi_dataset/data/dataset_catalog.json
  inflating: turtle/genome/md5sum.txt
Archive:  alligator_annotation.zip
  inflating: alligator/genome/README.md
  inflating: alligator/genome/ncbi_dataset/data/assembly_data_report.jsonl
  inflating: alligator/genome/ncbi_dataset/data/dataset_catalog.json
  inflating: alligator/genome/md5sum.txt
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % find turtle/genome
```

```
-type f
turtle/genome/ncbi_dataset/data/assembly_data_report.jsonl
turtle/genome/ncbi_dataset/data/GCF_011386835.1/protein.faa
turtle/genome/ncbi_dataset/data/GCF_011386835.1/genomic.gff
turtle/genome/ncbi_dataset/data/dataset_catalog.json
turtle/genome/README.md
turtle/genome/md5sum.txt
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % find alligator/geno
me -type f
alligator/genome/ncbi_dataset/data/assembly_data_report.jsonl
alligator/genome/ncbi_dataset/data/dataset_catalog.json
alligator/genome/README.md
alligator/genome/md5sum.txt
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
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(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % datasets download g
enome accession GCA_030867095.1 \
    --include gff3,protein \
    --filename alligator_annotation.zip
\ Collecting genome records    1 record
Downloading: alligator_annotation.zip    3.83kB valid data package
Validating package files [================================================] 100%
 4/4
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % unzip -o alligator_
annotation.zip -d alligator/genome
Archive:  alligator_annotation.zip
  inflating: alligator/genome/README.md
  inflating: alligator/genome/ncbi_dataset/data/assembly_data_report.jsonl
  inflating: alligator/genome/ncbi_dataset/data/dataset_catalog.json
  inflating: alligator/genome/md5sum.txt
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % find alligator/geno
me -type f
alligator/genome/ncbi_dataset/data/assembly_data_report.jsonl
alligator/genome/ncbi_dataset/data/dataset_catalog.json
alligator/genome/README.md
alligator/genome/md5sum.txt
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % nano notes/turtle_g
```

```
enes_loci.txt
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % mv notes/turtle_gen
es_loci.txt notes/turtle_genes_loci.csv
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % ls alligator/genome
gator_GCF_030867095.1_copy.gff  protein.faa
md5sum.txt                      README.md
ncbi_dataset
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % makeblastdb \
    -in alligator/genome/protein.faa \
    -dbtype prot \
    -out alligator_proteins \
    -parse_seqids


Building a new DB, current time: 09/27/2026 17:43:57
New DB name:   /Users/zainabsiddiqui/Downloads/Crocs/Data/synteny/alligator_prot
eins
New DB title:  alligator/genome/protein.faa
Sequence type: Protein
Keep MBits: T
Maximum file size: 3000000000B
Adding sequences from FASTA; added 48040 sequences in 0.951271 seconds.


(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
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(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % nano notes/turtle_f
lanking_ids.txt
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % datasets download g
ene gene-id \
    --inputfile turtle_flanking_ids.txt \
    --include protein \
    --filename turtle_flanking_proteins.zip
Error: 'open turtle_flanking_ids.txt: no such file or directory' opening input f
ile: 'turtle_flanking_ids.txt'

Use datasets download gene gene-id <command> --help for detailed help about a co
mmand.
```

```
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % datasets download g
ene gene-id \
    --inputfile notes/turtle_flanking_ids.txt \
    --include protein \
    --filename turtle_flanking_proteins.zip
Collecting 7 gene records [================================================] 100
% 7/7
Downloading: turtle_flanking_proteins.zip    8.97kB valid data package
Validating package files [================================================] 100%
 5/5
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % unzip -o turtle_fla
nking_proteins.zip -d turtle/turtle_flanking_proteins
Archive:  turtle_flanking_proteins.zip
  inflating: turtle/turtle_flanking_proteins/README.md
  inflating: turtle/turtle_flanking_proteins/ncbi_dataset/data/protein.faa
  inflating: turtle/turtle_flanking_proteins/ncbi_dataset/data/data_report.jsonl

  inflating: turtle/turtle_flanking_proteins/ncbi_dataset/data/dataset_catalog.j
son
  inflating: turtle/turtle_flanking_proteins/md5sum.txt
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % ls alligator_protei
ns*
alligator_proteins.pdb  alligator_proteins.pog  alligator_proteins.ptf
alligator_proteins.phr  alligator_proteins.pos  alligator_proteins.pto
alligator_proteins.pin  alligator_proteins.pot
alligator_proteins.pjs  alligator_proteins.psq
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % head -20 turtle/tur
tle_flanking_proteins/ncbi_dataset/data/protein.faa
>XP_065429985.1 ARHGEF1 [organism=Chrysemys picta bellii] [GeneID=101943841] [is
oform=X1]
MSGIGEIFHWTLPAISDPPPGTGGAQGAHPPRIPRTQRPLPHPPPRSGAAFPVEGGARRGTQVGPPRAGS
WEPPKGRMDLDEPYNGGPAPGAHSGSSVAIIGAEDEDFENDIEPTPEDHSSVFQSMVVVKSHPAYLMAFL
QHVVLQFDSCPVLCYLHVELIRKASPKEGKKLFLDFYHTFLDKAGLLRVHIPAQVQFELDRSRPELLPDE
TLRQFLNDIQNCQEPEISRQLEDFRSKRLMGMTPGEQELAELESYHTRDRWIREAKEKQLAELLLARLDE
MHPTISTDEEKSSAIFAAIVTYMKYLGVKTKLGESKKSKGNFFRKKMSGSRKPEEPPKPRKGFSLLDATR
WNRGESHNSDFRQSKVPEGEAEKAVASERKVLKLLERSESRGKGAGTPGLEPPGVSVTVNPPPGEGTDAE
LGSEPQCPVEQGEAPPSEQPPEDSAENERLPGRLGRSESLRVTERKRSQRGSAKGKQLRSRSDVDLEAAA
RAGELQERPPLGPGRPRPGGVTPEPGGAGGGEPPQSFLLPQSEETEPRVSELEPEPPTWRQRVPPDTLLR
```

```
LKKGEVKRQEVINELFLTEHAHGRMLRVLLEVFYQPLLSEGFFQETELANIFPGLDELMDVHTTFLESLK
KLREKSDCVIAEIGDTLLARFDGTEGSWFQKISARFCSRQSFALEQLKAKQRKETRFSQFIQEAESQPRC
RRLQLKDIIPIEMQRLTKYPLLLENIAKYTEEEEERAKVQRAAECCRQILNHVNQEVRHMENFMKLKDYQ
KRLDLSSLKQSTDPLLSEFKNTDISKRNLVHEGGLTWRVSKDKAVDVHVLLLDDILVLLQRQEERLVLRC
HSRPPGPTPEPRQVLSPIIKLSSAMTREVATDRNAFYVIFSWENGAQIYELVAQTVSERKNWCALISETA
GSLKLPGSNRQKPRRAAPSLSAPYSPSYYREPLLSTSENGNSSKERLHSDEREKDGALEGMEFEERSLEK
PEKDSERILAELLALHKPQAMGCEGGLAAAALERVSALKRLLVGGSPLPEDRDPSSPEDGWRRAPENRGA
GDEDEERDAGDGAESEGGAPGGAPNGPRLSGLGVAAPGRGGARLLHCADAPAVEGRAAQLPAPGGGCSQA
EGHGRRLFPAAAAPGQVQLPPLPPASHEGQPCRSPLAPEEEAARVFCPPAGGPPGGGRCWSVVPCLGARR
VYT
>XP_065429986.1 ARHGEF1 [organism=Chrysemys picta bellii] [GeneID=101943841] [is
oform=X2]
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % grep -c "^>" turtle
_flanking_proteins/ncbi_dataset/data/protein.faa
grep: turtle_flanking_proteins/ncbi_dataset/data/protein.faa: No such file or di
rectory
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % grep -c "^>" turtle
/turtle_flanking_proteins/ncbi_dataset/data/protein.faa
18
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
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(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % blastp \
    -query turtle/turtle_flanking_proteins.faa \
    -db alligator_proteins \
    -evalue 1e-10 \
    -outfmt "6 qseqid sseqid pident length evalue bitscore" \
    -out blast/flanking_genes_vs_alligator.tsv
Command line argument error: Argument "query". File is not accessible:  `turtle/
turtle_flanking_proteins.faa'
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
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(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % blastp \
```

```
    -query turtle/turtle_flanking_proteins/ncbi_dataset/data/protein.faa \
    -db alligator_proteins \
    -evalue 1e-10 \
    -outfmt "6 qseqid sseqid pident length evalue bitscore" \
    -out blast/flanking_genes_vs_alligator.tsv
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % head -n 10 blast/fl
anking_genes_vs_alligator.tsv
XP_065429985.1  ref|XP_059574825.1|     37.556  892     3.20e-142       469
XP_065429985.1  ref|XP_059574824.1|     37.556  892     5.43e-142       468
XP_065429985.1  ref|XP_059574823.1|     37.500  896     5.76e-142       468
XP_065429985.1  ref|XP_059574822.1|     37.500  896     6.38e-142       468
XP_065429985.1  ref|XP_059574819.1|     37.500  896     6.52e-142       469
XP_065429985.1  ref|XP_059574821.1|     37.500  896     7.56e-142       468
XP_065429985.1  ref|XP_059575649.1|     53.141  382     1.03e-119       406
XP_065429985.1  ref|XP_059575649.1|     44.186  215     3.02e-46        183
XP_065429985.1  ref|XP_059575648.1|     53.141  382     1.18e-119       406
XP_065429985.1  ref|XP_059575648.1|     44.186  215     3.16e-46        183
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
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(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % head -n 10 blast/al
ligator_tophits.csv
,sseqid_alt
0,XP_059574666.1
1,XP_059574124.1
2,XP_059574803.1
3,XP_059574144.1
4,XP_059574144.1
5,XP_059574654.1
6,XP_059574144.1
7,XP_059574144.1
8,XP_059574144.1
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % head -n 10 blast/al
ligator_tophits.csv
sseqid_alt
XP_059574666.1
XP_059574124.1
XP_059574803.1
XP_059574144.1
XP_059574144.1
XP_059574654.1
XP_059574144.1
XP_059574144.1
XP_059574144.1
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % head -n 10 blast/al
ligator_tophits.csv
sseqid_alt
XP_059574666.1
```

```
XP_059574124.1
XP_059574803.1
XP_059574144.1
XP_059574144.1
XP_059574654.1
XP_059574144.1
XP_059574144.1
XP_059574144.1
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % head -n 10 blast/al
ligator_tophits.csv
XP_059574666.1
XP_059574124.1
XP_059574803.1
XP_059574144.1
XP_059574144.1
XP_059574654.1
XP_059574144.1
XP_059574144.1
XP_059574144.1
XP_059574653.1
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % ls
alligator                       turtle
alligator_annotation.zip        turtle_annotation.zip
blast                           turtle_flanking_proteins.zip
notes                           turtle_prdm9.jsonl
regions                         turtle_prdm9.zip
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % nano blast/find_all
igator_coords.py
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % python blast/find_a
lligator_coords.py
Traceback (most recent call last):
  File "/Users/zainabsiddiqui/Downloads/Crocs/Data/synteny/blast/find_alligator_
coords.py", line 8, in <module>
    with open("alligator_tophits.txt") as f:
         ~~~~^^^^^^^^^^^^^^^^^^^^^^^^^
FileNotFoundError: [Errno 2] No such file or directory: 'alligator_tophits.txt'
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % nano blast/find_all
igator_coords.py
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
```

```
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % python blast/find_a
lligator_coords.py
Traceback (most recent call last):
  File "/Users/zainabsiddiqui/Downloads/Crocs/Data/synteny/blast/find_alligator_
coords.py", line 8, in <module>
    with open("alligator_tophits.csv") as f:
         ~~~~^^^^^^^^^^^^^^^^^^^^^^^^^
FileNotFoundError: [Errno 2] No such file or directory: 'alligator_tophits.csv'
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % nano blast/find_all
igator_coords.py
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % python blast/find_a
lligator_coords.py
Traceback (most recent call last):
  File "/Users/zainabsiddiqui/Downloads/Crocs/Data/synteny/blast/find_alligator_
coords.py", line 13, in <module>
    with open(gff_file, encoding="utf-8") as f:
         ~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^
FileNotFoundError: [Errno 2] No such file or directory: '../alligator/genome/gat
or_GCF_030867095.1_copy.gff'
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % nano blast/find_all
igator_coords.py
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % python blast/find_a
lligator_coords.py
Found 47 of 47 accessions.
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % grep "XP_059583279.
1" alligator/genome/gator_GCF_030867095.1_copy.gff | head
NC_081824.1     Gnomon  CDS     81197   81307   .       +       0       ID=cds-X
P_059583279.1;Parent=rna-XM_059727296.1;Dbxref=GeneID:102577176,GenBank:XP_05958
3279.1;Name=XP_059583279.1;gbkey=CDS;gene=LOC102577176;product=zinc finger prote
in 420 isoform X6;protein_id=XP_059583279.1
NC_081824.1     Gnomon  CDS     81391   82098   .       +       0       ID=cds-X
P_059583279.1;Parent=rna-XM_059727296.1;Dbxref=GeneID:102577176,GenBank:XP_05958
3279.1;Name=XP_059583279.1;gbkey=CDS;gene=LOC102577176;product=zinc finger prote
in 420 isoform X6;protein_id=XP_059583279.1
NC_081824.1     Gnomon  CDS     82691   82850   .       +       0       ID=cds-X
P_059583279.1;Parent=rna-XM_059727296.1;Dbxref=GeneID:102577176,GenBank:XP_05958
3279.1;Name=XP_059583279.1;gbkey=CDS;gene=LOC102577176;product=zinc finger prote
in 420 isoform X6;protein_id=XP_059583279.1
```

```
NC_081824.1     Gnomon  CDS     86313   86485   .       +       2       ID=cds-X
P_059583279.1;Parent=rna-XM_059727296.1;Dbxref=GeneID:102577176,GenBank:XP_05958
3279.1;Name=XP_059583279.1;gbkey=CDS;gene=LOC102577176;product=zinc finger prote
in 420 isoform X6;protein_id=XP_059583279.1
NC_081824.1     Gnomon  CDS     89508   89634   .       +       0       ID=cds-X
P_059583279.1;Parent=rna-XM_059727296.1;Dbxref=GeneID:102577176,GenBank:XP_05958
3279.1;Name=XP_059583279.1;gbkey=CDS;gene=LOC102577176;product=zinc finger prote
in 420 isoform X6;protein_id=XP_059583279.1
NC_081824.1     Gnomon  CDS     90241   90357   .       +       2       ID=cds-X
P_059583279.1;Parent=rna-XM_059727296.1;Dbxref=GeneID:102577176,GenBank:XP_05958
3279.1;Name=XP_059583279.1;gbkey=CDS;gene=LOC102577176;product=zinc finger prote
in 420 isoform X6;protein_id=XP_059583279.1
NC_081824.1     Gnomon  CDS     92589   94558   .       +       2       ID=cds-X
P_059583279.1;Parent=rna-XM_059727296.1;Dbxref=GeneID:102577176,GenBank:XP_05958
3279.1;Name=XP_059583279.1;gbkey=CDS;gene=LOC102577176;product=zinc finger prote
in 420 isoform X6;protein_id=XP_059583279.1
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % makeblastdb \
    -in alligator/genome/gator_NC_081838.1.fasta \
    -dbtype nucl \
    -out alligator_NC_081838.1_db/ \
    -parse_seqids


Building a new DB, current time: 10/05/2026 18:50:07
New DB name:   /Users/zainabsiddiqui/Downloads/Crocs/Data/synteny/alligator_NC_0
81838.1_db
New DB title:  alligator/genome/gator_NC_081838.1.fasta
Sequence type: Nucleotide
Keep MBits: T
Maximum file size: 3000000000B
Adding sequences from FASTA; added 1 sequences in 0.340225 seconds.


(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % awk '$3=="gene" &&
($9~/LOC102572676/ || $9~/ARHGEF1/)' alligator/genome/gator_GCF_030867095.1_copy
.gff
NC_081824.1     Gnomon  gene    42052365        42250070        .       +       I
D=gene-ARHGEF10;Dbxref=GeneID:102574027;Name=ARHGEF10;description=Rho guanine nu
cleotide exchange factor 10;gbkey=Gene;gene=ARHGEF10;gene_biotype=protein_coding
NC_081824.1     Gnomon  gene    467622559       467815228       .       -       I
D=gene-ARHGEF17;Dbxref=GeneID:102558628;Name=ARHGEF17;description=Rho guanine nu
cleotide exchange factor 17;gbkey=Gene;gene=ARHGEF17;gene_biotype=protein_coding
NC_081836.1     Gnomon  gene    3275713 3287703 .       -       .       ID=gene-
ARHGEF19;Dbxref=GeneID:109286291;Name=ARHGEF19;description=Rho guanine nucleotid
```

```
e exchange factor 19;gbkey=Gene;gene=ARHGEF19;gene_biotype=protein_coding
NC_081836.1     Gnomon  gene    3696582 3841025 .       +       .       ID=gene-
ARHGEF10L;Dbxref=GeneID:102575121;Name=ARHGEF10L;description=Rho guanine nucleot
ide exchange factor 10 like;gbkey=Gene;gene=ARHGEF10L;gene_biotype=protein_codin
g
NC_081836.1     Gnomon  gene    15708570        15732432        .       +       I
D=gene-ARHGEF16;Dbxref=GeneID:102565879;Name=ARHGEF16;description=Rho guanine nu
cleotide exchange factor 16;gbkey=Gene;gene=ARHGEF16;gene_biotype=protein_coding
NC_081838.1     Gnomon  gene    16532040        16571811        .       +       I
D=gene-ARHGEF11;Dbxref=GeneID:102571684;Name=ARHGEF11;description=Rho guanine nu
cleotide exchange factor 11;gbkey=Gene;gene=ARHGEF11;gene_biotype=protein_coding
NC_081838.1     Gnomon  gene    30944399        30948963        .       -       I
D=gene-ARHGEF1;Dbxref=GeneID:106737254;Name=ARHGEF1;description=Rho guanine nucl
eotide exchange factor 1;gbkey=Gene;gene=ARHGEF1;gene_biotype=protein_coding
NC_081838.1     Gnomon  gene    30960558        30963977        .       -       I
D=gene-LOC102572676;Dbxref=GeneID:102572676;Name=LOC102572676;description=toll-l
ike receptor 13;gbkey=Gene;gene=LOC102572676;gene_biotype=protein_coding
NC_081839.1     Gnomon  gene    30338673        30427527        .       -       I
D=gene-ARHGEF12;Dbxref=GeneID:102569441;Name=ARHGEF12;description=Rho guanine nu
cleotide exchange factor 12;gbkey=Gene;gene=ARHGEF12;gene_biotype=protein_coding
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % head -n 10 alligato
r/genome/gator_NC_081838.1.fasta
>NC_081838.1 Alligator mississippiensis isolate rAllMis1 chromosome 15, rAllMis1
, whole genome shotgun sequence
TCTAACCCTAACCTAAACCCTAACCTAACCTAACCCTAACCCTAACCCTAAACTAACCCTAACCCTAACC
CTAACCCTAACCCTAACCCTAACCCTAACCTACCTAACCCTAACCTAACCCATAACCCTAACCCTAACCC
TATCCCTACCCTAACCCTAACCCTAACCCTAACCCTAACCCTAACCCTAACCTAACCTACCTATCCCTAA
CCCTAACCCTAACCCTAACCCTAACCCTAACCCTACTAACCCTAACCCTAACCCTAACCCTAACCCTAAC
CCTAACCCTAACCCTAACCCTAACCCTAACCCTAACCCTAACCCTAACCCTAACCCTAACCCTAACCTAA
CCTAACCCTAACCCTAACCCTACCCTAACCCTAACCCTAACCCTAACCCTAACCCTAACCCTAACCCTAA
CCCTAACCCTACCTACCCTAACCCTACCTAAACCTAACCCTAAACCCTAACCCTAACCCTACCTAACCCT
AACCCTAACCCTAACCCTAACCCTAACCCTAACCCTAACCCTAACCCTAACCCTAACCTAACCCTAACCC
TAACCCTAACCCTAACCCTAACCCTAACCCTAACCCTAACCCTAACCCTAACCCTAACCCTAACCCTAAC
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % wc alligator/genome
/gator_NC_081838.1.fasta
  553389  553399 39290584 alligator/genome/gator_NC_081838.1.fasta
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
```

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(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % makeblastdb \
    -in regions/gator/LOC_to_ARHG.fa \
    -dbtype nucl \
    -out blast/alligator_LOCtoARHG__db/ \
    -parse_seqids


Building a new DB, current time: 10/06/2026 21:33:58
New DB name:   /Users/zainabsiddiqui/Downloads/Crocs/Data/synteny/blast/alligato
r_LOCtoARHG__db
New DB title:  regions/gator/LOC_to_ARHG.fa
Sequence type: Nucleotide
Keep MBits: T
Maximum file size: 3000000000B
Adding sequences from FASTA; added 1 sequences in 0.00282407 seconds.


(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % blastn \
    -query turtle/prdm9/ncbi_dataset/data/gene.fna \
    -db blast/alligator_LOCtoARHG_db \
    -evalue 1e-10 \
    -outfmt "6 qseqid sseqid pident length evalue bitscore" \
    -out blast/alligator_LOCtoARHG_db/LOCtoARHG_blasthits.csv
BLAST Database error: No alias or index file found for nucleotide database [blas
t/alligator_LOCtoARHG_db] in search path [/Users/zainabsiddiqui/Downloads/Crocs/
Data/synteny::]. Please verify the spelling of the BLAST database and its molecu
le type.
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % blastn \
    -query turtle/prdm9/ncbi_dataset/data/gene.fna \
    -db blast/alligator_LOCtoARHG_db/alligator_LOCtoARHG \
    -evalue 1e-10 \
    -outfmt "6 qseqid sseqid pident length evalue bitscore" \
    -out blast/alligator_LOCtoARHG_db/LOCtoARHG_blasthits.csv
BLAST Database error: No alias or index file found for nucleotide database [blas
t/alligator_LOCtoARHG_db/alligator_LOCtoARHG] in search path [/Users/zainabsiddi
qui/Downloads/Crocs/Data/synteny::]. Please verify the spelling of the BLAST dat
abase and its molecule type.
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % blastn \
    -query turtle/prdm9/ncbi_dataset/data/gene.fna \
    -db blast/alligator_LOCtoARHG_db/alligator_LOCtoARHG \
    -evalue 1e-10 \
    -outfmt "6 qseqid sseqid pident length evalue bitscore" \
    -out blast/alligator_LOCtoARHG_db/LOCtoARHG_blasthits.csv
BLAST Database error: File /Users/zainabsiddiqui/Downloads/Crocs/Data/synteny/bl
```

```
ast/alligator_LOCtoARHG_db/alligator_LOCtoARHG__db.ndb not found. If you renamed
 any BLAST database files, please use original file names, and makeblastdb to re
name the database. If you deleted any BLAST database files, you need to recreate
 the database.
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % blastn \
    -query turtle/prdm9/ncbi_dataset/data/gene.fna \
    -db blast/alligator_LOCtoARHG_db/alligator_LOCtoARHG \
    -evalue 1e-10 \
    -outfmt "6 qseqid sseqid pident length evalue bitscore" \
    -out blast/alligator_LOCtoARHG_db/LOCtoARHG_blasthits.csv
BLAST Database error: No alias or index file found for nucleotide database [blas
t/alligator_LOCtoARHG_db/alligator_LOCtoARHG] in search path [/Users/zainabsiddi
qui/Downloads/Crocs/Data/synteny::]. Please verify the spelling of the BLAST dat
abase and its molecule type.
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % blastn \
    -query turtle/prdm9/ncbi_dataset/data/gene.fna \
    -db blast/alligator_LOCtoARHG_db/alligator_LOCtoARHG \
    -evalue 1e-10 \
    -outfmt "6 qseqid sseqid pident length evalue bitscore" \
    -out blast/alligator_LOCtoARHG_db/LOCtoARHG_blasthits.csv
BLAST Database error: File /Users/zainabsiddiqui/Downloads/Crocs/Data/synteny/bl
ast/alligator_LOCtoARHG_db/alligator_LOCtoARHG__db.ndb not found. If you renamed
 any BLAST database files, please use original file names, and makeblastdb to re
name the database. If you deleted any BLAST database files, you need to recreate
 the database.
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % makeblastdb \
    -in regions/gator/LOC_to_ARHG.fa \
    -dbtype nucl \
    -out blast/alligator_LOCtoARHG_db/ \
    -parse_seqids


Building a new DB, current time: 10/06/2026 21:46:43
New DB name:   /Users/zainabsiddiqui/Downloads/Crocs/Data/synteny/blast/alligato
r_LOCtoARHG_db
New DB title:  regions/gator/LOC_to_ARHG.fa
Sequence type: Nucleotide
Keep MBits: T
Maximum file size: 3000000000B
Adding sequences from FASTA; added 1 sequences in 0.00357699 seconds.


(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
```

```
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % blastn \
    -query turtle/prdm9/ncbi_dataset/data/gene.fna \
    -db blast/alligator_LOCtoARHG_db/ \
    -evalue 1e-10 \
    -outfmt "6 qseqid sseqid pident length evalue bitscore" \
    -out blast/alligator_LOCtoARHG_db/LOCtoARHG_blasthits.csv
BLAST Database error: No alias or index file found for nucleotide database [blas
t/alligator_LOCtoARHG_db/] in search path [/Users/zainabsiddiqui/Downloads/Crocs
/Data/synteny::]. Please verify the spelling of the BLAST database and its molec
ule type.
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % blastn \
    -query turtle/prdm9/ncbi_dataset/data/gene.fna \
    -db blast/alligator_LOCtoARHG_db/ \
    -evalue 1e-10 \
    -outfmt "6 qseqid sseqid pident length evalue bitscore" \
    -out blast/alligator_LOCtoARHG_db/LOCtoARHG_blasthits.csv
BLAST Database error: No alias or index file found for nucleotide database [blas
t/alligator_LOCtoARHG_db/] in search path [/Users/zainabsiddiqui/Downloads/Crocs
/Data/synteny::]. Please verify the spelling of the BLAST database and its molec
ule type.
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % blastn \
    -query turtle/prdm9/ncbi_dataset/data/gene.fna \
    -db blast/alligator_LOCtoARHG_db \
    -evalue 1e-10 \
    -outfmt "6 qseqid sseqid pident length evalue bitscore" \
    -out blast/alligator_LOCtoARHG_db/LOCtoARHG_blasthits.csv
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % rm blast/alligator_
LOCtoARHG_db/LOCtoARHG_blasthits.csv
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % blastn \
    -query turtle/prdm9/ncbi_dataset/data/gene.fna \
    -db blast/alligator_LOCtoARHG_db \
    -evalue 1e-10 \
    -outfmt "6 qseqid sseqid pident length evalue bitscore" \
    -out blast/alligator_LOCtoARHG_db/LOCtoARHG_blasthits.csv
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % head -n 5 turtle/pr
dm9/ncbi_dataset/data/gene.fna
>NC_088810.1:992989-1017260 PRDM9 [organism=Chrysemys picta bellii] [GeneID=1019
```

```
47848] [chromosome=20]
GGGGCGGGAATGCCCCGCCCTTCTTGTCTAGGGGCGGGGCTGTTTCTTTCCCTCTCCTCTCTCCTCGCCT
CGCGCCGCGGCCGTTAGCCGCCTTCTGCCGGGCGCGAGACCAGCGGCTCTCTTTCCCGCGCTGCCCGCTG
ATTGGCGGCTACCGTCCCTTCGCCCCACGCTGATTGGCCACAATGTACCAATGCCCCTCGATGATTGGCT
AGAGGTTTTGGCGGGCTCTGGTAGACGAAGGCGAAGGGTTGTTGGGGTAGAGGCTGGTGCAGCCCAATTC
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % blastn \
    -query turtle/prdm9/ncbi_dataset/data/gene.fna \
    -db blast/alligator_LOCtoARHG_db \
    -evalue 1e-6 \
    -outfmt "6 qseqid sseqid pident length evalue bitscore" \
    -out blast/alligator_LOCtoARHG_db/LOCtoARHG_blasthits.csv
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % blastn \
    -query turtle/prdm9/ncbi_dataset/data/gene.fna \
    -db blast/alligator_LOCtoARHG_db \
    -evalue 1e-3 \
    -outfmt "6 qseqid sseqid pident length evalue bitscore" \
    -out blast/alligator_LOCtoARHG_db/LOCtoARHG_blasthits.csv
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % blastn \
    -query turtle/prdm9/ncbi_dataset/data/gene.fna \
    -db blast/alligator_LOCtoARHG_db \
    -evalue 1e-3 \
    -outfmt "6 qseqid sseqid pident length evalue bitscore" \
    -out blast/alligator_LOCtoARHG_db/LOCtoARHG_blasthits.csv
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % blastn \
    -query turtle/prdm9/ncbi_dataset/data/gene.fna \
    -db blast/alligator_LOCtoARHG_db \
    -evalue 1e-6 \ -task dc-megablast \
    -outfmt "6 qseqid sseqid pident length evalue bitscore" \
    -out blast/alligator_LOCtoARHG_db/LOCtoARHG_blasthits.csv
USAGE
  blastn [-h] [-help] [-import_search_strategy filename]
    [-export_search_strategy filename] [-task task_name] [-db database_name]
    [-dbsize num_letters] [-gilist filename] [-seqidlist filename]
    [-negative_gilist filename] [-negative_seqidlist filename]
    [-taxids taxids] [-negative_taxids taxids] [-taxidlist filename]
    [-negative_taxidlist filename] [-no_taxid_expansion]
    [-entrez_query entrez_query] [-db_soft_mask filtering_algorithm]
    [-db_hard_mask filtering_algorithm] [-subject subject_input_file]
    [-subject_loc range] [-query input_file] [-out output_file]
    [-evalue evalue] [-word_size int_value] [-gapopen open_penalty]
    [-gapextend extend_penalty] [-perc_identity float_value]
    [-qcov_hsp_perc float_value] [-max_hsps int_value]
    [-xdrop_ungap float_value] [-xdrop_gap float_value]
    [-xdrop_gap_final float_value] [-searchsp int_value] [-penalty penalty]
    [-reward reward] [-no_greedy] [-min_raw_gapped_score int_value]
    [-template_type type] [-template_length int_value] [-dust DUST_options]
    [-filtering_db filtering_database]
    [-window_masker_taxid window_masker_taxid]
```

```
    [-window_masker_db window_masker_db] [-soft_masking soft_masking]
    [-ungapped] [-culling_limit int_value] [-best_hit_overhang float_value]
    [-best_hit_score_edge float_value] [-subject_besthit]
    [-window_size int_value] [-off_diagonal_range int_value]
    [-use_index boolean] [-index_name string] [-lcase_masking]
    [-query_loc range] [-strand strand] [-parse_deflines] [-outfmt format]
    [-show_gis] [-num_descriptions int_value] [-num_alignments int_value]
    [-line_length line_length] [-html] [-sorthits sort_hits]
    [-sorthsps sort_hsps] [-max_target_seqs num_sequences]
    [-num_threads int_value] [-mt_mode int_value] [-remote] [-version]

DESCRIPTION
   Nucleotide-Nucleotide BLAST 2.17.0+

Use '-help' to print detailed descriptions of command line arguments
===================================================================

Error: Too many positional arguments (1), the offending value:  -task
Error:  (CArgException::eSynopsis) Too many positional arguments (1), the offend
ing value:  -task
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % blastn \
    -query turtle/prdm9/ncbi_dataset/data/gene.fna \
    -db blast/alligator_LOCtoARHG_db \
    -evalue 1e-6  -task dc-megablast \
    -outfmt "6 qseqid sseqid pident length evalue bitscore" \
    -out blast/alligator_LOCtoARHG_db/LOCtoARHG_blasthits.csv
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % head -n 5 regions/g
ator/LOC_to_ARHG.fa
>NC_081838.1_LOC102572676_to_ARHGEF1 Extracted region from 30944399 to 30963977
GGAAAGCAGGACTTTATTGGCAGCAAGATCTCCCCCCCCACCCCCGCATAAAAACTCCATT
AAAACCTCTGATATATATATATATAAAAAGTCTCCAGCAAAAGGCTGACCCCCATGGCCCCG
ATCCAGATGCAGAGCTGGGGGGAACTGAGGGGTATCGGCTCCAGGGACCTAAGAGAGAGG
AGAGGAGAGGAAAGGGTCAAGCCCCACCTTGCCCCCAGCCTGGCCCCCCCCAGAGCCCTC
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % head -n 5 /Users/za
inabsiddiqui/Downloads/Crocs/Data/synteny/turtle/prdm9/ncbi_dataset/data/protein
.faa
>XP_042703928.2 PRDM9 [organism=Chrysemys picta bellii] [GeneID=101947848] [isof
```

```
orm=X2]
MSDNAGEGRASEKRKEQKGRKAGESSKVEDEFKELSIYFPKEQWAEMGAWEKTRYRNMKRNYEFMVELGL
PTPKPAFMCRGRRPQNPIIYDSESDEDWTPKPLVKTFRPPRKSGSLKQEKKKRQNISQNEKTVLEKASEN
QTDTEKLDKNSDVGTRIPVTNVVDLKTDLPRLAASKTQGQSTQVNTYSLRKREKKVYVEINEPQDDDYLF
CEICLTFFIDECSVHGPPVFIKDPIVETGQERRAALTLPPGMRIGLSSIPQAGLGVWNEADTLPIGVHFG
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % python3 -c "
from Bio import SeqIO
from Bio.Seq import Seq
with open('/Users/zainabsiddiqui/Downloads/Crocs/Data/synteny/alligator/genome/g
ator_NC_081838.1_protein.fasta', 'w') as out:
    for record in SeqIO.parse('/Users/zainabsiddiqui/Downloads/Crocs/Data/synten
y/alligator/genome/gator_NC_081838.1.fasta', 'fasta'):
        prot = record.seq.translate(to_stop=True)
        out.write(f'>{record.id}\n{prot}\n')
"
<frozen importlib._bootstrap>:491: RuntimeWarning: The global interpreter lock (
GIL) has been enabled to load module 'Bio.Align._aligncore', which has not decla
red that it can run safely without the GIL. To override this behavior and keep t
he GIL disabled (at your own risk), run with PYTHON_GIL=0 or -Xgil=0.
/opt/anaconda3/envs/prdm9_synteny/lib/python3.14t/site-packages/Bio/Seq.py:2874:
 BiopythonWarning: Partial codon, len(sequence) not a multiple of three. Explici
tly trim the sequence or add trailing N before translation. This may become an e
rror in future.
  warnings.warn(
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % wc alligator/genome
/gator_NC_081838.1.fasta
  553389  553399 39290584 alligator/genome/gator_NC_081838.1.fasta
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % head -n5 /Users/zai
nabsiddiqui/Downloads/Crocs/Data/synteny/alligator/genome/gator_NC_081838.1_prot
ein.fasta
>NC_081838.1
SNPNLNPNLT
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % wc /Users/zainabsid
diqui/Downloads/Crocs/Data/synteny/alligator/genome/gator_NC_081838.1_protein.fa
sta
       2       2      24 /Users/zainabsiddiqui/Downloads/Crocs/Data/synteny/alli
gator/genome/gator_NC_081838.1_protein.fasta
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
```

```
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % makeblastdb \
    -in regions/gator/LOC_to_ARHG_protein.fa \
    -dbtype prot \
    -out blast/alligator_LOCtoARHG_protein_db/alligator_LOCtoARHG_protein_db \

    -parse_seqids
USAGE
  makeblastdb [-h] [-help] [-help-full] [-in input_file] [-input_type type]
    -dbtype molecule_type [-title database_title] [-parse_seqids]
    [-hash_index] [-mask_data mask_data_files] [-mask_id mask_algo_ids]
    [-mask_desc mask_algo_descriptions] [-gi_mask]
    [-gi_mask_name gi_based_mask_names] [-out database_name]
    [-blastdb_version version] [-max_file_sz number_of_bytes]
    [-metadata_output_prefix ] [-logfile File_Name] [-taxid TaxID]
    [-taxid_map TaxIDMapFile] [-oid_masks oid_masks] [-version]

DESCRIPTION
   Application to create BLAST databases, version 2.17.0+

Use '-help' to print detailed descriptions of command line arguments
============================================================================

Error: Too many positional arguments (1), the offending value:
Error:  (CArgException::eSynopsis) Too many positional arguments (1), the offend
ing value:
zsh: command not found: -dbtype
zsh: command not found: -out
zsh: command not found: -parse_seqids
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % makeblastdb \
    -in regions/gator/LOC_to_ARHG_protein.fa \
    -dbtype prot \
    -out blast/alligator_LOCtoARHG_protein_db/ \

    -parse_seqids
USAGE
  makeblastdb [-h] [-help] [-help-full] [-in input_file] [-input_type type]
    -dbtype molecule_type [-title database_title] [-parse_seqids]
    [-hash_index] [-mask_data mask_data_files] [-mask_id mask_algo_ids]
    [-mask_desc mask_algo_descriptions] [-gi_mask]
    [-gi_mask_name gi_based_mask_names] [-out database_name]
    [-blastdb_version version] [-max_file_sz number_of_bytes]
```

```
    [-metadata_output_prefix ] [-logfile File_Name] [-taxid TaxID]
    [-taxid_map TaxIDMapFile] [-oid_masks oid_masks] [-version]

DESCRIPTION
   Application to create BLAST databases, version 2.17.0+

Use '-help' to print detailed descriptions of command line arguments
==========================================================================

Error: Too many positional arguments (1), the offending value:
Error:  (CArgException::eSynopsis) Too many positional arguments (1), the offend
ing value:
zsh: command not found: -dbtype
zsh: command not found: -out
zsh: command not found: -parse_seqids
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % makeblastdb \
    -in regions/gator/LOC_to_ARHG_protein.fa \
    -dbtype prot \
    -out alligator_LOCtoARHG_protein \
    -parse_seqids
USAGE
  makeblastdb [-h] [-help] [-help-full] [-in input_file] [-input_type type]
    -dbtype molecule_type [-title database_title] [-parse_seqids]
    [-hash_index] [-mask_data mask_data_files] [-mask_id mask_algo_ids]
    [-mask_desc mask_algo_descriptions] [-gi_mask]
    [-gi_mask_name gi_based_mask_names] [-out database_name]
    [-blastdb_version version] [-max_file_sz number_of_bytes]
    [-metadata_output_prefix ] [-logfile File_Name] [-taxid TaxID]
    [-taxid_map TaxIDMapFile] [-oid_masks oid_masks] [-version]

DESCRIPTION
   Application to create BLAST databases, version 2.17.0+

Use '-help' to print detailed descriptions of command line arguments
==========================================================================

Error: Too many positional arguments (1), the offending value:
Error:  (CArgException::eSynopsis) Too many positional arguments (1), the offend
ing value:
zsh: command not found: -dbtype
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % makeblastdb -in reg
ions/gator/LOC_to_ARHG_protein.fa -dbtype prot -out blast/alligator_LOCtoARHG_pr
otein_db/alligator_protein_db -parse_seqids
```

```
Building a new DB, current time: 10/08/2026 12:37:29
New DB name:   /Users/zainabsiddiqui/Downloads/Crocs/Data/synteny/blast/alligato
r_LOCtoARHG_protein_db/alligator_protein_db
New DB title:  regions/gator/LOC_to_ARHG_protein.fa
Sequence type: Protein
Keep MBits: T
Maximum file size: 3000000000B
Adding sequences from FASTA; added 1 sequences in 0.00405192 seconds.


(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % blastp \
    -query turtle/prdm9/ncbi_dataset/data/gene_protein.fna \
    -db blast/alligator_LOCtoARHG_protein_db \
    -evalue 1e-10 \
    -outfmt "6 qseqid sseqid pident length evalue bitscore" \
    -out blast/alligator_LOCtoARHG_protein_db/blastp_LOCtoARHG_hits.csv
BLAST Database error: No alias or index file found for protein database [blast/a
lligator_LOCtoARHG_protein_db] in search path [/Users/zainabsiddiqui/Downloads/C
rocs/Data/synteny::]. Please verify the spelling of the BLAST database and its m
olecule type.
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % blastp \
    -query turtle/prdm9/ncbi_dataset/data/gene_protein.fna \
    -db blast/alligator_LOCtoARHG_protein_db/ \
    -evalue 1e-10 \
    -outfmt "6 qseqid sseqid pident length evalue bitscore" \
    -out blast/alligator_LOCtoARHG_protein_db/blastp_LOCtoARHG_hits.csv
BLAST Database error: No alias or index file found for protein database [blast/a
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(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % blastn \
    -query turtle/prdm9/ncbi_dataset/data/gene.fna \
    -db blast/alligator_LOCtoARHG_db/alligator_LOCtoARHG_db \
    -evalue 1e-3  \
    -outfmt "6 qseqid sseqid pident length evalue bitscore" \
    -out blast/alligator_LOCtoARHG_db/LOCtoARHG_blasthits.csv
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
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(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % blastx \
    -query turtle/prdm9/ncbi_dataset/data/gene.fna \
    -db blast/alligator_LOCtoARHG_db/alligator_LOCtoARHG_db \
    -evalue 1e-3  \
    -outfmt "6 qseqid sseqid pident length evalue bitscore" \
    -out blast/alligator_LOCtoARHG_db/LOCtoARHG_blasthits.csv
```

```
BLAST Database error: No alias or index file found for protein database [blast/a
lligator_LOCtoARHG_db/alligator_LOCtoARHG_db] in search path [/Users/zainabsiddi
qui/Downloads/Crocs/Data/synteny::]. Please verify the spelling of the BLAST dat
abase and its molecule type.
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
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(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % blastx \
    -query turtle/prdm9/ncbi_dataset/data/gene.fna \
    -db blast/alligator_LOCtoARHG_protein_db/alligator_protein_db \
    -evalue 1e-3  \
    -outfmt "6 qseqid sseqid pident length evalue bitscore" \
    -out blast/alligator_LOCtoARHG_protein_db/LOCtoARHG_blastx_hits.csv
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(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % nano blast/alligato
r_LOCtoARHG_protein_db/LOCtoARHG_blastx_hits.csv
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(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % tblastx \
    -query turtle/prdm9/ncbi_dataset/data/gene.fna \
    -db blast/alligator_LOCtoARHG_db/alligator_LOCtoARHG_db \
    -evalue 1e-3  \
    -outfmt "6 qseqid sseqid pident length evalue bitscore" \
    -out blast/alligator_LOCtoARHG_db/LOCtoARHG_blastx_hits.csv
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
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(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % tblastx \
    -query turtle/prdm9/ncbi_dataset/data/gene.fna \
    -db blast/alligator_LOCtoARHG_db/alligator_LOCtoARHG_db \
    -evalue 1e-3  \
    -outfmt 6 \
    -out blast/alligator_LOCtoARHG_db/LOCtoARHG_blastx_hits.csv
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
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(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % tblastx \
    -query turtle/prdm9/ncbi_dataset/data/gene.fna \
    -db blast/alligator_LOCtoARHG_db/alligator_LOCtoARHG_db \
    -evalue 1e-3  \
    -outfmt "6 qseqid   sseqid  pident  length  mismatch        gapopen qstart
qend    sstart  send    qframe    sframe    evalue      bitscore" \
    -out blast/alligator_LOCtoARHG_db/LOCtoARHG_blastx_hits.csv
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % tblastx \
    -query turtle/prdm9/ncbi_dataset/data/gene.fna \
    -db blast/alligator_LOCtoARHG_db/alligator_LOCtoARHG_db \
    -evalue 1e-3  \
    -outfmt 6 \
    -out blast/alligator_LOCtoARHG_db/LOCtoARHG_blastx_hits.csv
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(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % tblastx \
    -query turtle/prdm9/ncbi_dataset/data/gene.fna \
    -db blast/alligator_LOCtoARHG_db/alligator_LOCtoARHG_db \
    -evalue 1e-3  \
    -outfmt 0 \
    -out blast/alligator_LOCtoARHG_db/LOCtoARHG_tblastx_alignment.txt
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
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(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % grep ">" turtle/prd
m9/ncbi_dataset/data/gene.fna
>NC_088810.1:992989-1017260 PRDM9 [organism=Chrysemys picta bellii] [GeneID=1019
47848] [chromosome=20]
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
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tlePRDM9.txt
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
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(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % cat regions/gator/L
```

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OC_to_ARHG.fa >> LOCtoARHG_turtlePRDM9.txt
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % head -n 5 LOCtoARHG
_turtlePRDM9.txt
>NC_081838.1_LOC102572676_to_ARHGEF1 Extracted region from 30944399 to 30963977
GGAAAGCAGGACTTTATTGGCAGCAAGATCTCCCCCCCACCCCCGCATAAAAACTCCATT
AAAACCTCTGATATATATATATATAAAAAGTCTCCAGCAAAAGGCTGACCCCCATGGCCCCG
ATCCAGATGCAGAGCTGGGGGGAACTGAGGGGTATCGGCTCCAGGGACCTAAGAGAGAGG
AGAGGAGAGGAAAGGGTCAAGCCCCACCTTGCCCCCAGCCTGGCCCCCCCCCAGAGCCCTC
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % cat turtle/prdm9/nc
bi_dataset/data/gene.fna >> LOCtoARHG_turtlePRDM9.txt
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % grep ">" LOCtoARHG_
turtlePRDM9.txt
>NC_081838.1_LOC102572676_to_ARHGEF1 Extracted region from 30944399 to 30963977
>NC_088810.1:992989-1017260 PRDM9 [organism=Chrysemys picta bellii] [GeneID=1019
47848] [chromosome=20]
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
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(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % makeblastdb -in tib
erius/LOCtoARHG_Tiberius_g2.fasta -dbtype prot -out blast/tiberius_g2_db/tiberiu
sg2 -parse_seqids


Building a new DB, current time: 10/13/2026 11:30:59
New DB name:   /Users/zainabsiddiqui/Downloads/Crocs/Data/synteny/blast/tiberius
_g2_db/tiberiusg2
New DB title:  tiberius/LOCtoARHG_Tiberius_g2.fasta
Sequence type: Protein
Keep MBits: T
Maximum file size: 3000000000B
Adding sequences from FASTA; added 1 sequences in 0.00348997 seconds.


(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
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(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny %
(prdm9_synteny) zainabsiddiqui@Zainabs-MacBook-Pro synteny % makeblastdb -in tib
erius/LOCtoARHG_Tiberius_g2.fasta -dbtype nucl -out blast/tiberius_g2_db/tiberiu
sg2 -parse_seqids


Building a new DB, current time: 10/13/2026 11:31:55
New DB name:   /Users/zainabsiddiqui/Downloads/Crocs/Data/synteny/blast/tiberius
_g2_db/tiberiusg2
New DB title:  tiberius/LOCtoARHG_Tiberius_g2.fasta
```